

Non-Hermitian TDVP/Arnoldi Study in 05_arnoldi

Project Benchmark Report

March 13, 2026

1 Scope and Goal

This report documents all benchmark experiments implemented and run in 05_arnoldi. The central goal is to evaluate MPS-based time evolution for non-Hermitian linear ODEs,

$$\frac{d}{dt}\psi(t) = K\psi(t), \quad (1)$$

against exact dense full-Hilbert-space references on small systems, and to compare algorithmic choices (Arnoldi vs Lanczos, TDVP vs BUG variants).

2 Experiment Families

2.1 Nearest-neighbor non-Hermitian model

Model:

$$K = -iJ \sum_{j=1}^{N-1} Z_j Z_{j+1} - ig \sum_{j=1}^N X_j - \gamma \sum_{j=1}^N \frac{I - Z_j}{2}. \quad (2)$$

Default parameters used:

$$N = 10, \quad J = 1.0, \quad g = 0.7, \quad \gamma = 0.2, \quad dt = 0.05, \quad t_{\max} = 1.0, \quad D \in \{4, 8, 16\}.$$

2.2 Long-range non-Hermitian model

Model:

$$K_{\text{lr}} = -i \sum_{1 \leq i < j \leq N} J_{ij} Z_i Z_j - ig \sum_{j=1}^N X_j - \gamma \sum_{j=1}^N \frac{I - Z_j}{2}, \quad (3)$$

with

$$J_{ij} = J_0 \lambda^{|i-j|-1}. \quad (4)$$

Default runs used

$$N = 10, \quad J_0 = 1.0, \quad \lambda = 0.5, \quad g = 0.7, \quad \gamma = 0.2, \quad dt = 0.05, \quad t_{\max} = 1.0, \quad D \in \{4, 8, 16, 32\}.$$

Additional sweeps included $\lambda \in \{0.25, 0.5, 0.75\}$ for BUG-vs-TDVP comparisons.

2.3 Exact reference

For all benchmarks, the reference is dense full-space evolution:

$$U = \exp(dt K), \quad \psi_{n+1} = U\psi_n.$$

No tensor-train compression is used in the reference.

3 Implementation Notes

3.1 TDVP entry point and Krylov switch

The repository’s existing routines were used:

- `Yaq.s.Algorithms.two_site_tdvp!`
- `Yaq.s.Algorithms.set_krylov_ishermitian_mode!`

Because TDVP internally applies local propagators of the form $\exp(-i \, dt \, A)$, the experiment passes

$$A = iK$$

to realize target dynamics $\exp(dt \, K)$.

3.2 Krylov mode behavior check

Direct test results:

- Forced `:lanczos` does *not* auto-switch and stays in Lanczos mode.
- `:auto` on the non-Hermitian problem switches to Arnoldi.

Therefore, using `--krylov lanczos` explicitly forces a method that is not appropriate for this non-Hermitian case.

4 Key Results

4.1 Nearest-neighbor, Arnoldi TDVP

From `results/summary.txt`:

Bond dim D	Final state error $\ \psi_{\text{tdvp}} - \psi_{\text{ref}}\ _2$	Final max local error $\max_j \Delta\langle Z_j \rangle $
4	4.68×10^{-4}	5.79×10^{-6}
8	4.59×10^{-6}	2.69×10^{-7}
16	4.59×10^{-6}	2.69×10^{-7}

Accuracy saturates already around $D = 8$ for this setup.

4.2 Long-range, Arnoldi TDVP ($\lambda = 0.5$)

From `results.longrange/summary.txt`:

Bond dim D	Final state error	Final max local error
4	4.86×10^{-2}	3.94×10^{-4}
8	3.31×10^{-3}	2.50×10^{-5}
16	5.67×10^{-4}	3.24×10^{-5}
32	5.66×10^{-4}	3.24×10^{-5}

Long-range dynamics are harder, but Arnoldi TDVP remains accurate and converges with D .

4.3 Arnoldi vs Lanczos for non-Hermitian evolution

Observed behavior:

- Arnoldi: stable and accurate.
- Lanczos (forced): large persistent errors ($\sim 10^{-1}$ to 10^0 scale), many non-Hermitian warnings.

Conclusion: for this non-Hermitian class, Arnoldi is clearly the correct Krylov choice.

4.4 TDVP vs BUG (second-order variants)

Comparison runs in `results_bug2_compare/summary.txt` show:

- TDVP (Arnoldi) improves strongly with D across nearest-neighbor and long-range cases.
- BUG-based trajectories (both original and non-Hermitian-adapted variant tested) stayed at much larger errors and did not improve meaningfully with D in this benchmark suite.

5 Selected Plots

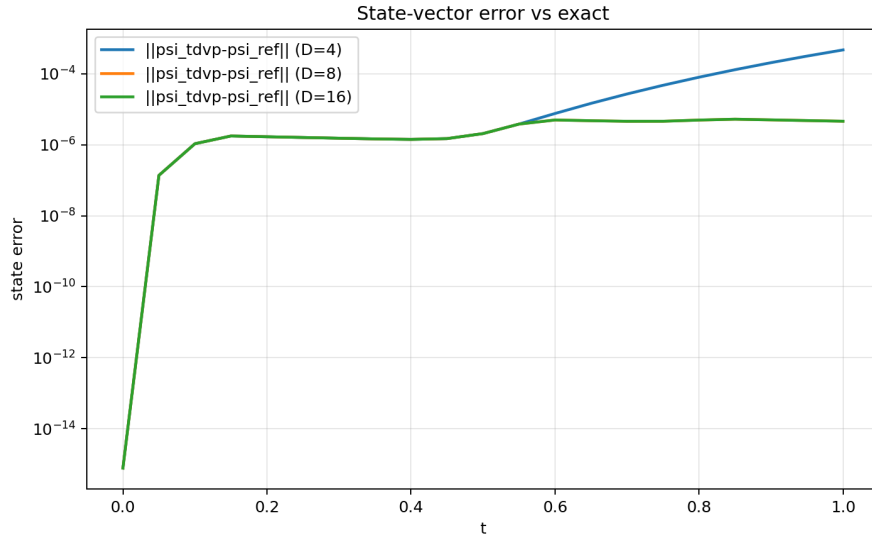


Figure 1: Nearest-neighbor benchmark: TDVP state error vs exact reference.

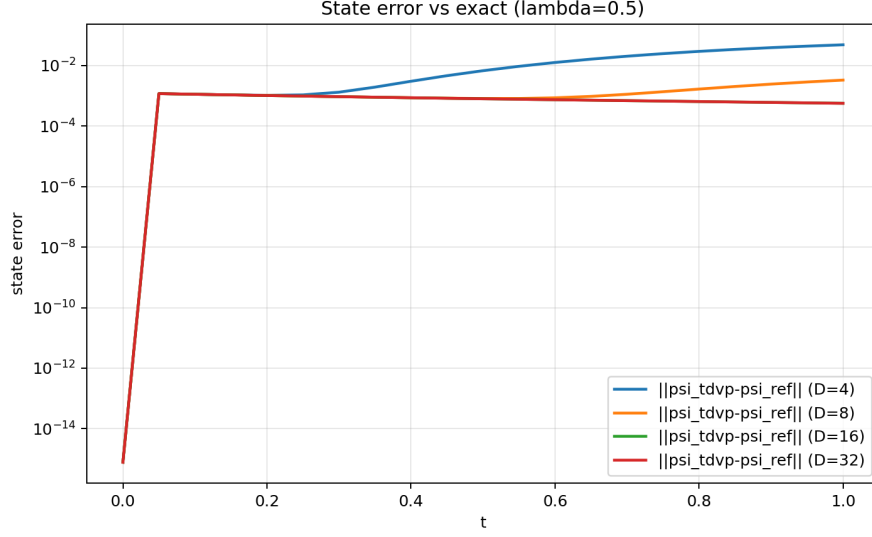
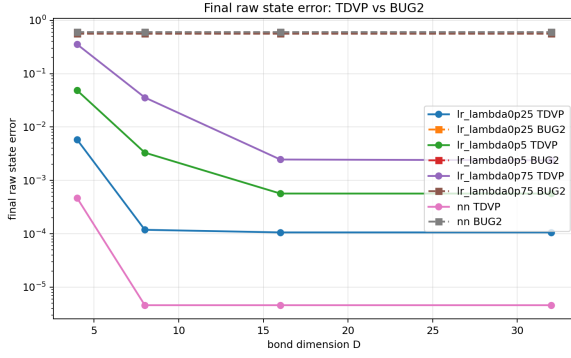
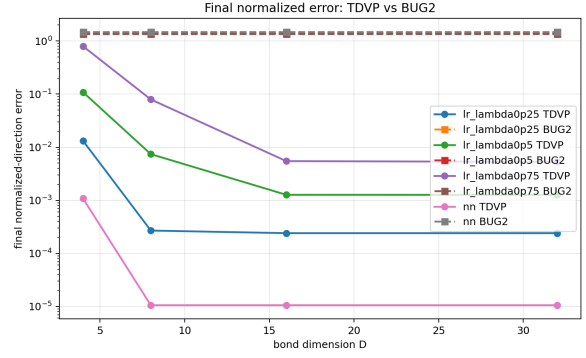


Figure 2: Long-range benchmark ($\lambda = 0.5$): TDVP state error vs exact reference.



(a) Raw final state error



(b) Normalized-direction final error

Figure 3: TDVP vs BUG comparisons across models and bond dimensions.

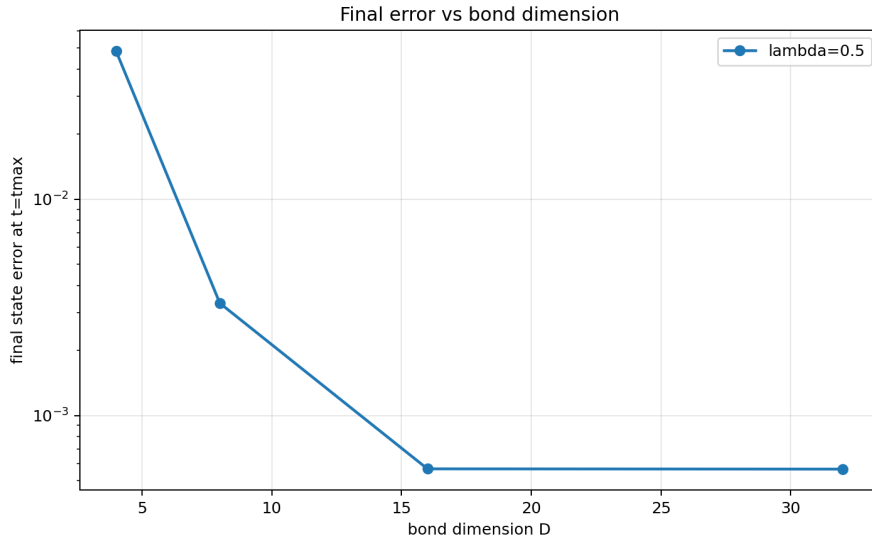


Figure 4: Long-range TDVP: final error vs bond dimension D .

6 Conclusions

- The benchmark infrastructure in `05_arnoldi` successfully validates non-Hermitian TDVP against exact dense references.
- Arnoldi is the preferred Krylov mode for these non-Hermitian operators.
- Forced Lanczos is not suitable in this setting and should not be used for production non-Hermitian runs.
- In this benchmark suite, BUG variants were consistently inferior to Arnoldi-TDVP.

7 Outlook / Next Steps

Potential follow-up directions:

1. **Higher-system-size scaling:** increase N while replacing dense exact reference by controlled high- D reference windows to extend the regime.
2. **Adaptive tolerances:** tune Krylov tolerance and truncation thresholds jointly, report accuracy/runtime Pareto fronts.
3. **Observable-focused validation:** add nonlocal observables (two-point correlators, structure factors) in addition to local $\langle Z_j \rangle$.
4. **Non-Hermitian stress tests:** sweep γ , λ , and stronger long-range couplings; map where a given D ceases to be reliable.
5. **Method development branch:** if BUG remains of interest, prototype a truly non-Hermitian-targeted BUG variant (without implicit normalization effects) in an isolated branch and benchmark again against the same suite.